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Boron Based Fluoride Anion Receptors: Electrochemical and Sensory Applications

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Dedicated to Professor Russell N. Grimes on the occasion of his 80th Birthday

Abstract

Boron based fluoride anion receptors as electrolyte additives would significantly improve lithium ion battery performance. Fluorinated versions of triarylboranes, boronate esters, boroxines, and other boron based anion receptors have potential applications in the design of high specific energy lithium ion batteries. There is also an emerging interest in recent years in the design of colorimetric sensors for the fluoride anion, especially in aqueous systems, using boron based anion receptors. This mini review highlights the recent trends in the design and applications of the boron based fluoride anion fluoride anion receptors as electrolyte additives and as colorimetric sensors.

Key words: Anion receptors; fluoride anion receptor; boroxines; lithium ion batteries; colorimetric sensors

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1. Introduction

Boron-based anion receptors such as tris(1,1,1,3,3,3-hexafluoropropan-2-yl) borate (TPFB), tris(pentafluorophenyl) borate (THFIPB), tris(pentafluorophenyl)borane (TPFPB) were first demonstrated by Yang, McBreen and coworkers as fluoride, oxide or peroxide anion specific receptors when used as additives in lithium ion batteries (based on LiF, Li₂O or Li₂O₂ as electrolytes in carbonate-based solvents) (Figure 1).¹⁻⁴ The state of the art electrolyte materials such as LiPF₆ exhibit relatively low lithium ion transference numbers, and therefore there is renewed interest to develop selective anion receptors that would facilitate enhanced dissociation of the lithium salts, thereby resulting in increased lithium ion transference numbers.⁵ The dual ion intercalation secondary rechargeable lithium ion batteries using lithium fluoride salts are particularly attractive in this regard for achieving relatively high specific energy values. However, the relative insolubility of the LiF in many nonaqueous electrolyte solvents limits its use as a viable electrolyte salt, and therefore the use of the fluoride specific anion receptors as electrolyte additives is a clear alternative towards achieving enhanced solubility of LiF and thereby the relatively increased lithium ion transference numbers.⁵⁻¹¹

The selective reversible binding of these boron-based anion receptors resulted in enhanced lithium ion transference numbers. For example, in the presence of 0.4 M TPFPB in 0.2 M Li₂O/1:1 EC:DMC (ethylene carbonate and dimethyl carbonate solvents) the lithium ion transference number was observed to be as high as 0.89. Similarly, a lithium ion transference number of 0.74 was observed for 0.4 M TPFPB in 0.2 M Li₂O₂/1:1 PC (propylene carbonate):DMC. These Li⁺ transference numbers are typically 100% higher than the values for the commonly used electrolytes in the absence of the anion receptors.² The enhanced Li⁺ transference numbers would result in relatively high lithium ion conductivities in lithium ion cells.

The use of these anion receptors allows the use of LiF as a low molecular weight electrolyte in the lithium ion batteries. Whereas LiF is essentially insoluble in the nonaqueous electrolyte solvents such as dimethyl ether (DME) and carbonate based solvents, in the presence of anion receptors LiF is relatively highly soluble, and typically 1 M solutions of LiF in dimethyl ether could be obtained so that high lithium ion conductivities could be achieved in lithium ion batteries.^{1, 3, 4}

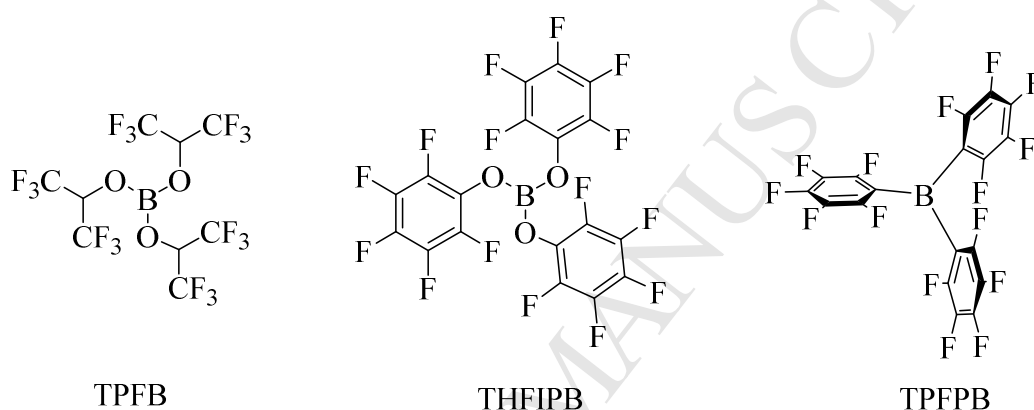


Figure 1. Structures of a trialkoxyborane TPFB, a triaryloxyborane THIPB, and a perfluorinated triarylborane TPFPB

Whereas there are relatively highly selective colorimetric fluoride anion receptors based on Bronsted acid–base reactions or hydrogen bonding interactions as in the case of urea-,¹²⁻¹⁵ indole-,¹⁶ pyrrole-,^{17, 18} carbazole-,¹⁹ thiourea-²⁰ and calixarene-based²¹⁻²⁴ anion receptors (Figure 2), the boron-based anion receptors have superior electrochemical and thermal stabilities, and thereby are compatible with the state of the art carbonate-based nonaqueous electrolytes ($E_{ox} > 4.5$ V vs Li/Li⁺).²⁵ The borate- and boroxine-based fluoride anion receptors are readily accessible from the corresponding relatively inexpensive boric acid derivatives. Furthermore, the fluoride anion affinities of the boron based anion receptors could be reliably predicted by ab initio calculations. Amine and coworkers' ab initio calculations, for example, show that the

fluoride anion affinity is reduced by the introduction of the alkoxy groups on the boron center and that the fluoride anion affinity increases with the substitution of the electron-withdrawing groups on the boron (e.g., fluoroalkyl or fluoroaryl aryl substituents on the boron).²⁶ The anion receptor TPFPB was shown also to complex with PF_6^- anion (i.e., in addition to its complexation with the fluoride anion) so that the lithium ion transference number (and thereby the ionic conductivity) of the cells consisting of LiPF_6 electrolyte are significantly increased.²⁷

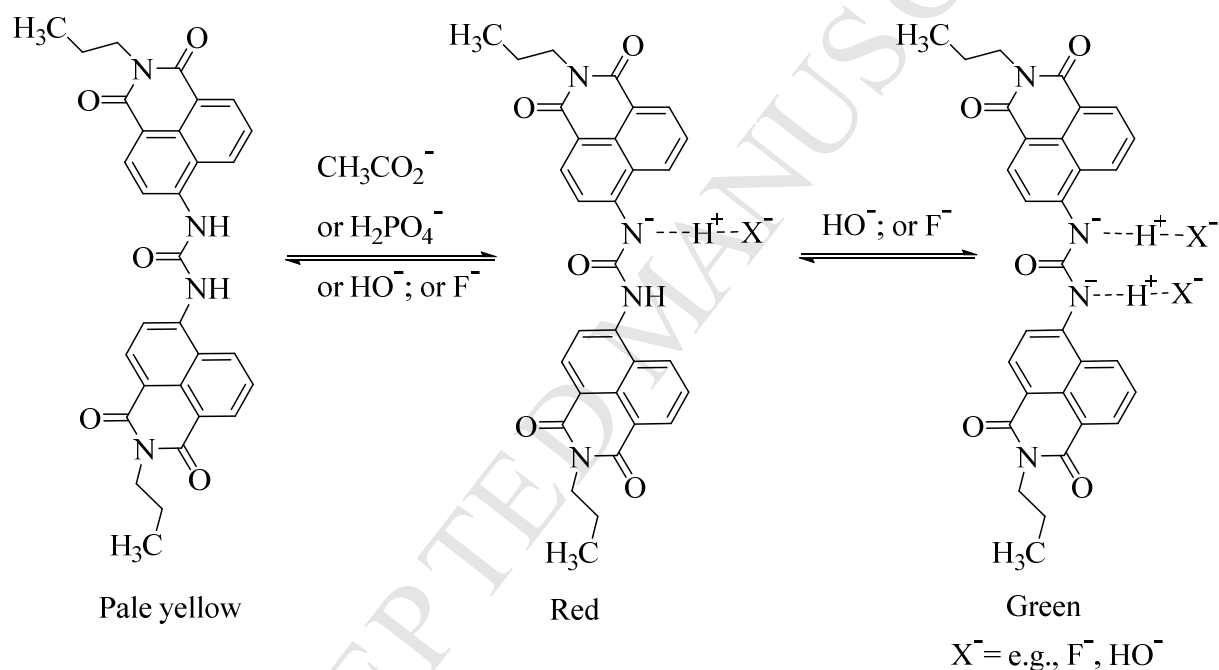


Figure 2. Color changes upon complexation of fluoride anion by a chromogenic urea derivative; the NH bond of the urea derivative is successively, almost completely deprotonated upon addition of the tetrabutylammonium fluoride.¹³ The second deprotonation step could be achieved using the fluoride or hydroxide anions, whereas other anions such as acetate and dihydrogen phosphate could only achieve the first deprotonation.

The applications of these anion receptors and the associated electrolytes in lithium ion batteries and lithium-air batteries has been reviewed,^{7, 25} and therefore we will only briefly

address the fundamental concepts involved in their applications in lithium ion batteries and highlight the recent applications of the fluoride anion receptors. Appropriately selective fluoride anion receptors have potential impact on the commercial applications of the lithium ion and metal-air batteries.

Due to the environmental as well as toxicity effects of fluoride anions there has been renewed interest in developing analytical techniques for the detection and quantification of the fluoride ion concentrations in aqueous and nonaqueous systems. Fluoride anion selective colorimetric sensors, in particular in aqueous systems, are highly desirable as many industrial pollutants release fluoride ions in to the waste water systems. The commonly used Bronsted acid–base sensors involving urea and other amines (such as pyrrole, thiourea and guanidinium based compound)- based anion receptors are particularly useful for these applications. However, most of these Bronsted acid-based anion receptors suffer from their relatively non-selective nature towards the fluoride anion, as they also form acid-base complexes with various other anions such as acetate, dihydrogen phosphate, and other halides, exhibiting color changes. Boron-based anion receptors (Lewis acid–base complex based) are especially well suited for the selective analysis of the fluoride anion as boron has relatively superior fluoride anion affinity due to the relatively high B-F bond strengths. However, in aqueous solvents, due to the strong solvation of the fluoride anion by water, selective fluoride anion binding by the anion receptors is still a challenge. This area of fluorescence and colorimetric detection and quantitative analysis of the fluoride anions has been well reviewed, and therefore we highlight only the most illustrative recent examples of the boron-based colorimetric fluoride anion sensors.²⁸⁻³⁶ Gabbai and coworkers have recently reported sterically crowded triarylboranes as fluoride anion specific receptors (*vide infra*).³⁷⁻⁴⁰ In the latter cases, complementarity of the hydrogen-bonding

interactions from the neighboring N-H or O-H bonds significantly enhances the fluoride anion binding (*vide infra*). A wide variety of the sterically crowded triarylboranes have been synthesized recently and investigated for their fluoride anion binding and for their potential use as colorimetric sensors (*vide infra*).

2. Cyclic boronate esters as electrolyte additives in lithium ion batteries

The sterically less encumbered cyclic boronate esters are expected to have relatively higher fluoride anion affinities in nonaqueous solvents as compared to those of their acyclic analogues. McBreen and coworkers have synthesized the perfluorinated catechol- and perfluorinated pinacol- derived borate esters of the pentafluorophenylboronic acid (4,5,6,7-tetrafluoro-2-(2,3,4,5,6)pentafluorophenyl) borzodioxaborole (**1**) and 2-(pentafluorophenyl)-4,4,5,5-tetrakis-(trifluoromethyl)-1,3,2-dioxaborolane (**2**) (Figure 3)) through the reactions of the pentafluorophenylboronic acid with the corresponding tetrafluoro catechol and perfluoropinacol, respectively, and explored their effectiveness in lithium ion battery performance.³ The cyclic boronate esters significantly enhanced the ionic conductivity of LiF salt dissolved in ethyl carbonate-dimethyl carbonate (EC-EMC; 1:2) solvent. The catechol and pinacol derived borate esters (**1** and **2**) have electrochemical stability windows of 4.05 (4.43)⁴¹ and 5.1 V vs Li⁺/Li, respectively.³ Amine and coworkers have demonstrated the use of compound **1** as a bifunctional additive for overcharge protection and for improving cell cycle performance, of lithium ion cells.⁴¹

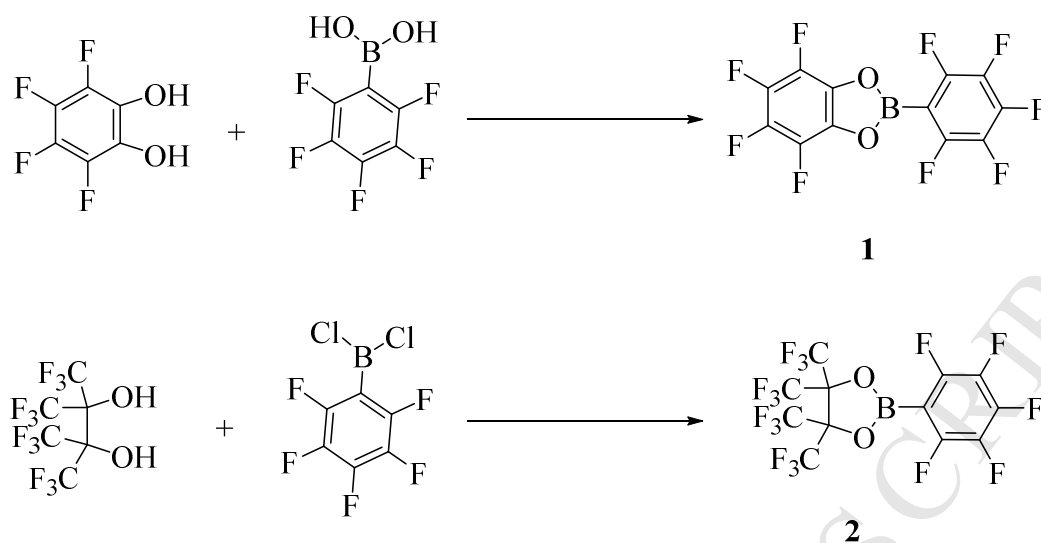


Figure 3. Synthesis of the cyclic boronate esters **1** and **2**

Perfluoroalkyl substituted cyclic boronate esters have a dramatic impact on the capacity retention of the lithium ion cells. For example, the lithium ion cells (graphite-based negative electrode and high voltage $\text{Li}_{1.2}\text{Ni}_{0.15}\text{Mn}_{0.55}\text{Co}_{0.1}\text{O}_2$ -based positive electrodes using 3:7 EC/DMC + 1.2 M LiPF_6) consisting of perfluorooctylethylene glycol ester of phenylboronic acid (4-perfluorooctyl-2-phenyl-1,3,2-dioxaborolane; PFO-PBE) display a much lower capacity fade of 21%, as compared to those of the ethylene glycol ester of the phenylboronic acid (2-phenyl-1,3,2-dioxaborolane; PBE) that result in a capacity fade of 34%.⁴² This improved capacity retention is also associated with reduced impedance rise as compared to the cells lacking the boronate ester additives. The latter boronic acid esters, PBE and PFO-PBE, are readily synthesized in one step through dehydrative dimerization of the phenylboronic acid with the corresponding 1,2-diols (Figure 4).

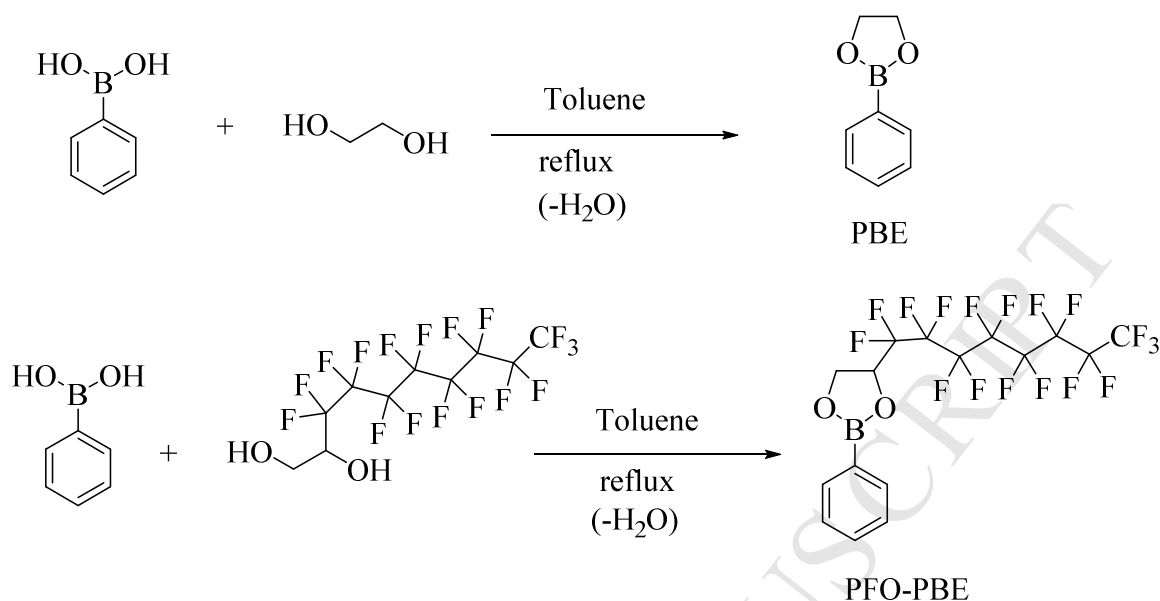


Figure 4. Synthesis of the cyclic boronate esters PBE and PFO-PBE

3. Boroxines as fluoride anion receptors in lithium ion batteries

Boroxines are the trimeric cyclic borates, obtained readily by thermal dehydration of the corresponding boronic acids. The electrophilicity and therefore the anion receptor binding efficiency can be modulated by the number of the fluorines on the aryl rings, and thus are well suited as electrolyte additives in lithium ion batteries, especially those that involve the use of low molecular weight electrolyte LiF, toward the design of high specific energy batteries. In order to be effective anion receptor in lithium ion or metal air batteries, the binding constants of fluoride anion to the receptor should be appropriately optimized so that the binding is sufficiently reversible. Boroxine based anion receptors have such advantage since the anion binding efficiency is modulated by the electronic and steric effects of the substituents around the boron. The fluoride anion binding by the boroxines could be modulated over a wide range by varying the number of the neighboring fluorines, as shown by density functional theory computations.¹¹

Fluorinated boroxines, such as tris(2,6-difluorophenyl)boroxine (DF), tris(2,4,6-trifluorophenyl)boroxine (TF), and tris(pentafluorophenyl)boroxine (PF) were synthesized by the dehydrative trimerization of the corresponding fluorinated arylboronic acids (Figure 5).¹¹ In addition to the fluoride anion binding, the latter boroxines also form noncovalent adducts with relatively polar solvents such as acetonitrile, dimethylsulfoxide and ethyl acetate, as shown by their ¹⁹F NMR spectra, whereas such solvent effects are relatively unimportant in nonpolar solvents such as CH₂Cl₂. The sensitivity of these boroxines to the solvent binding increases with the increase in the number of the aryl fluorines. That is, the perfluorinated boroxine PF is relatively more solvated as compared to the DF and TF. Similarly, the fluoride anion binding constants increase with the number of the aryl fluorines, as also substantiated by the ab initio calculations on the variously fluorinated boroxines.¹¹ The lithium ion conductivity of the electrolyte systems consisting of LiF electrolyte in propylene carbonate (PC) solvent is significantly increased when PC is used as additive,¹¹ and is comparable to that when tris(pentafluorophenyl)borane (TPFPB)^{2, 8, 10, 27, 43-50} is used as the electrolyte additive.

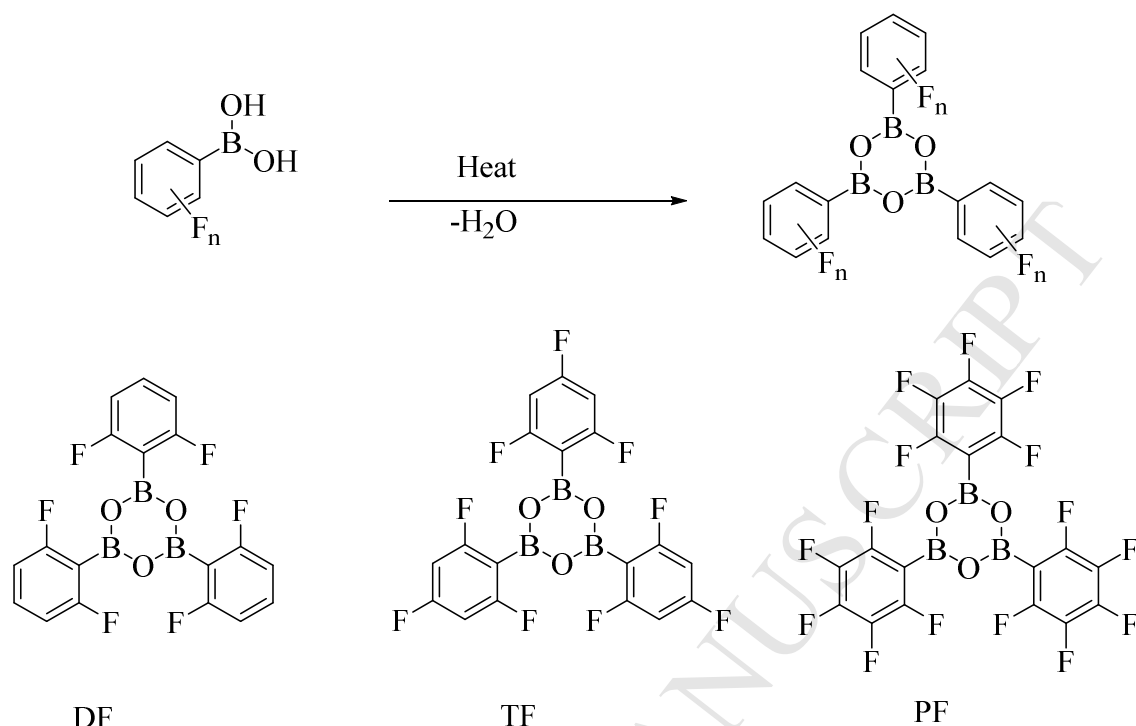


Figure 5. Synthesis of boroxines and structures of the fluorinated boroxines DF, TF, and PF

The ^{19}F and ^{11}B NMR spectra show that there is a rapid equilibration of the unsymmetrical fluoride adducts at ambient temperature through intramolecular fluoride migration, with an estimated barrier of $12.0 - 12.5 \text{ kcal mol}^{-1}$ derived through combined ^{19}F NMR and density functional theory (DFT) calculations (Figure 3). The rapid equilibration of the unsymmetrical B-F adducts is reflected by the observation of identical $\delta^{19}\text{F}$ for all the *ortho*-aryl fluorines and the disappearance of the ^{19}F signal for the boron-bound fluorine. However, the $\delta^{11}\text{B}$ of the fluorine-bound boron is distinct from the other two borons (implying that the rate of fluoride migration is sufficiently slow on the ^{11}B NMR time scale), and is significantly shielded due to the localized negative charge on the boron, which further supports the existence of the unsymmetrical boroxine-fluoride adduct. The high level density functional calculations are in accordance with this observation, the equilibration occurring through the relatively high energy symmetrical structure as a transition state (Figure 6).

Jobs' plots obtained through UV-vis spectroscopic studies showed the existence of a 1:1 boroxine-fluoride anion complex for the tris(2,6-difluorophenyl)boroxine, DF. However, in case of the tris(perfluorophenyl)boroxine (PF), in addition to the 1:1 boroxine-fluoride anion complex, multiple equilibria involving more than one fluoride adduct could be inferred through the UV-vis Job's plots. Although multiple equilibria are indicated for the PF-F⁻ adducts in the solution phase, PF shows a 1:1 adduct with fluoride anion in the gas phase, as indicated by its high resolution negative ion detected electrospray ionization time-of-flight mass spectrum. Thus, a mixture of the tris(perfluorophenyl)boroxine (PF) and KF shows a molecular ion peak at m/z 600.9835 (calculated m/z 600.9871) for the 1:1 boroxine-fluoride adduct.⁵¹

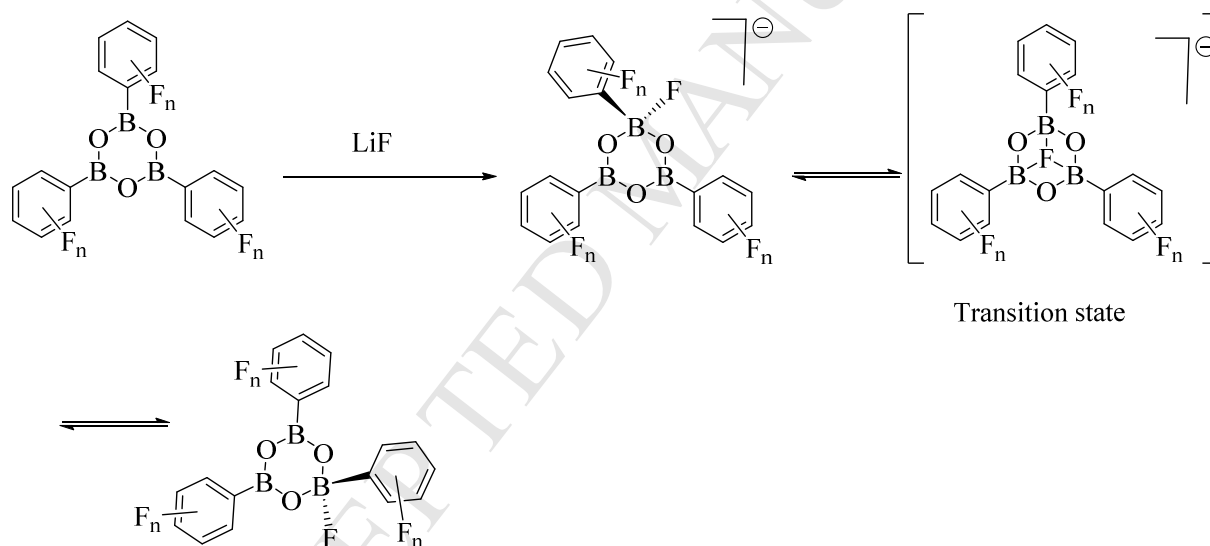
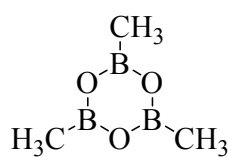


Figure 6. Equilibration of the unsymmetrically bound fluoride adducts of boroxines; experimental and theoretical estimates of their interconversion barriers is 12.0 – 12.5 kcal mol⁻¹.

It has been recently reported that trimethylboroxine (TMB; **3**), when used as an electrolyte additive, improves the lithium ion cell performance. In LiCoO₂/graphite lithium ion cells using ethylene carbonate ethylene carbonate (EC)-based electrolyte, 0.5% by weight of trimethylboroxine improved the capacity retention to 66% as compared to that of 29% in its absence.⁵²



3 (TMB)

This improved lithium ion cell performance in the presence of the boroxine **3** was attributed to the preferential oxidation of the boroxine **3** over the EC-based electrolyte at the positive electrode (LiCoO₂), which results in the formation of a protective film on the electrode surface. Similar improvements in the cycling performance and relatively high capacity retention was also observed in lithium ion cells consisting of LiCoPO₄/C cathodes in the presence of 0.5-1% of the TMB.^{53, 54}

4. Triarylboranes as colorimetric sensors of fluoride anion

In spite of the highly selective fluoride anion binding by the boroxines and other triaryl boranes and borates in polar aprotic solvents,⁶ there are virtually no selective boron-based fluoride anion receptors in aqueous medium. This is partly due to the high solvation of the fluoride anion by the water molecules, which decreases the nucleophilicity of the fluoride anion towards the boron. On the other hand, it has been elegantly demonstrated that certain strongly electrophilic, sterically crowded triarylborane ammonium and phosphonium salts (compounds **4-7**) are selective fluoride anion receptors in aqueous media (Figure 7).^{37, 38, 40, 55}

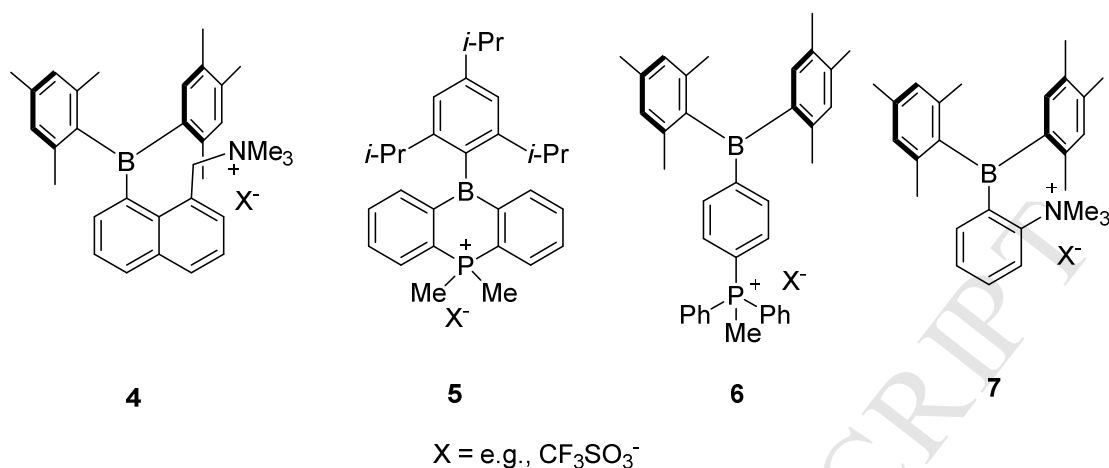


Figure 7. Structures of some of the sterically crowded triarylborane-based fluoride anion receptors; these boranes exhibit high fluoride anion affinity in aqueous systems.

The latter triarylborane ammonium and phosphonium salts have remarkably high fluoride anion binding constants in aqueous medium ($K \sim 10^3$ to 10^8 M^{-1}), and some of these compounds could also be used as phase transfer agents to transport fluoride anion from aqueous to organic medium.

Gabbai and coworkers have shown that *o*-(dimesitylboryl)trifluoroacetanilide (**8**) binds to fluoride anion in organic solvents, such as tetrahydrofuran (THF) and acetone, with a binding constant of 10^8 M^{-1} , which is two orders of magnitude greater than that for the trimesitylborane (Figure 8).³⁹ The relatively enhanced binding affinity in this compound is attributed to the additional stabilization provided by hydrogen bond formation through the amide hydrogen. The ^{19}F NMR of the fluoride complex of the latter borane shows a broad signal at $\delta^{19}\text{F} -164.4$, which is characteristic of related triarylborane–fluoride complexes. Further, the amide hydrogen appears as a doublet ($^1J_{\text{H-F}} = 36 \text{ Hz}$) at $\delta^1\text{H} 11.43$, confirming the presence of the hydrogen-bonding interaction with the boron-bound fluorine, and also indicating a 1:1 stoichiometry for the borane–fluoride anion complex. Similarly, sterically crowded

triarylboranes appended by an *o*-carborane moiety in one of the aryl rings (or all three aryl rings) show high fluoride anion affinity to the boron, due to the increase in the Lewis acidity of the boron, and they serve as potential colorimetric sensors for the fluoride anion.^{56, 57}

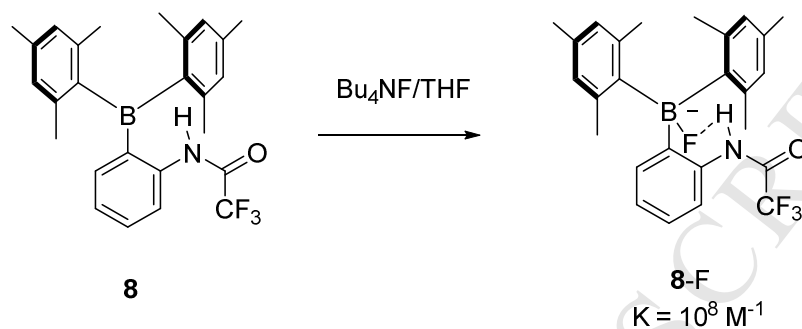


Figure 8. Augmentation of the fluoride anion binding to the boron due to the hydrogen-bonding interactions

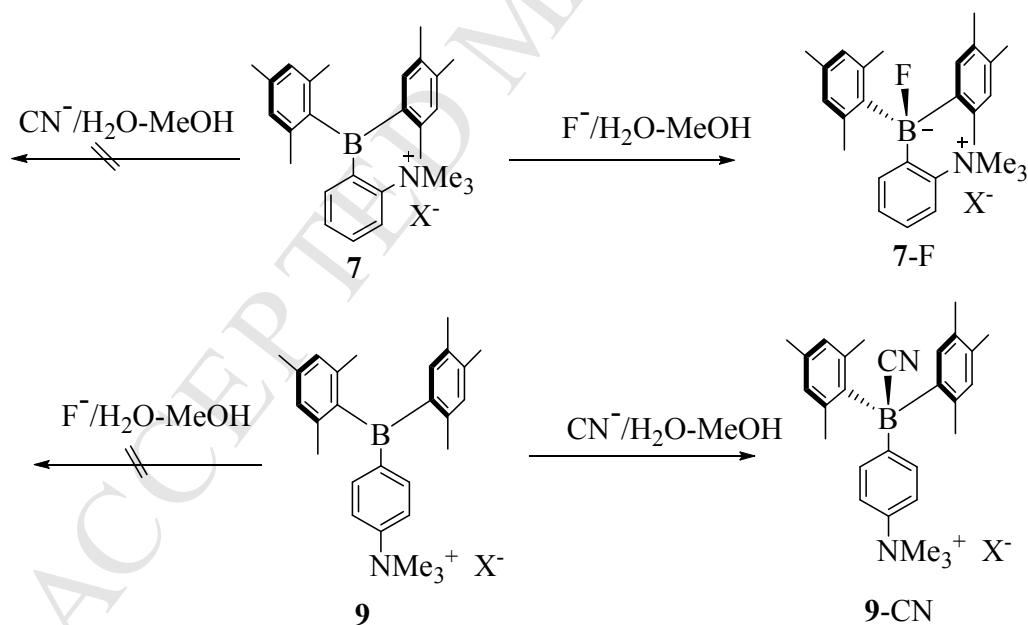


Figure 9. Fluoride and cyanide anion specific boron-based anion receptors; The complementary and high selectivity of these compounds to the fluoride and cyanide anions is ascribed to the combined steric and electronic effects.³⁸

Interestingly the *ortho*-trimethylammonium substituted dimesityl-derived borane **7** specifically binds to the fluoride anion and does not bind to the cyanide anion in aqueous methanol, whereas its *para*-trimethylammonium substituted analogue (compound **9**) binds specifically to the cyanide anion but not to the fluoride anion (Figure 9).³⁸ Computational studies show that this dichotomy arises due to the combined steric and electronic effects. The sterically encumbered compound **7**, preferentially binds to the relatively less sterically crowded fluoride anion and not to the relatively sterically crowded cyanide anion, whereas the corresponding *para*- isomer **9** binds to the cyanide anion as it lacks as much steric crowding at the boron center. On the other hand, the relatively lower coulombic effect afforded by the distant cationic site is not sufficient to allow complexation of the relatively more efficiently hydrated fluoride (and thus relatively less basic) anion in aqueous medium.³⁸

Neighboring indole and phenol moieties enhance binding of fluoride to boron in case of the triarylboranes. For example, the fluoride binding constants for the triaryl boranes having a phenol or an indole moiety attached to the boron (**10** and **11**, respectively) are an order of magnitude higher than those for the triarylboranes lacking such substituents (Figure 10).⁵⁸ Proton NMR studies showed that in addition to the significant covalent bonding to the boron center, the B–F bond in the fluoride adduct is stabilized by the hydrogen-bonding interactions with the neighboring indolyl N–H (**11**-F) and the phenolic O–H (**10**-F) bonds. Upon titration of the triarylboranes with tetrabutylammonium fluoride the $\delta^1\text{H}$ for the phenolic O–H is shifted downfield from $\delta^1\text{H}$ 7.22 to $\delta^1\text{H}$ 8.85, and the OH absorption appears as a doublet with a coupling constant ($^1J_{\text{H-F}}$) of 45 Hz, indicating the hydrogen-bonding interaction with the B–F fluorine. Although similar coupling to the indolyl N–H bond was not observed in case of the fluoride adduct of the indolyl-substituted triarylborane, the significant downfield shifted $\delta^1\text{H}$ for

the indolyl N–H proton indicates its relatively weaker hydrogen bond with the B–F fluorine.⁵⁸

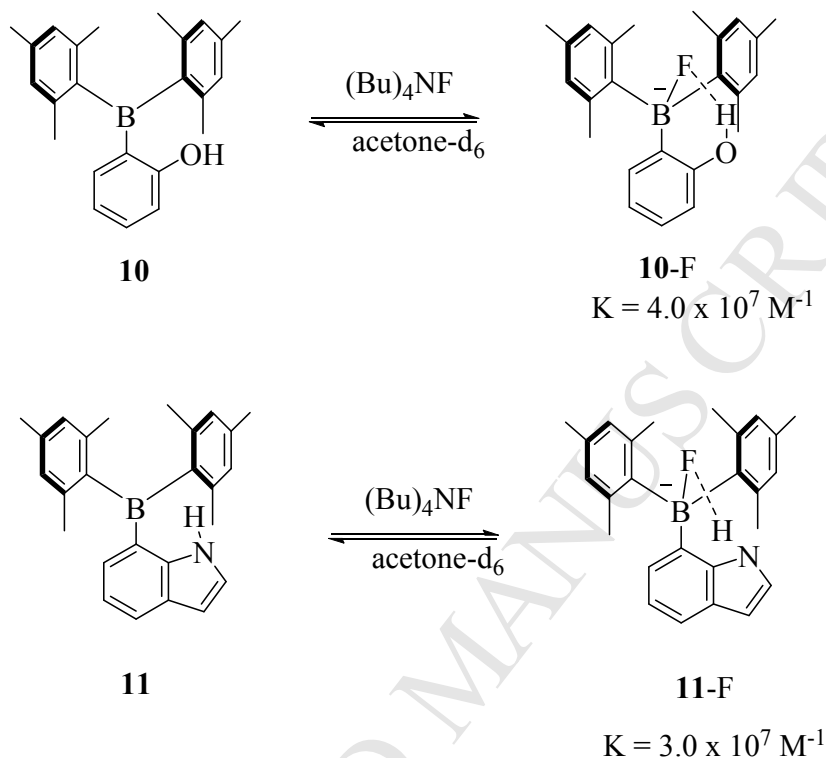


Figure 10. Augmentation of the fluoride anion binding to boron due to the neighboring hydroxyl and amino groups

A concave (Λ) shaped molecule, consisting of Troger's base as the linker connecting a sterically crowded triarylborane and dimethylaminophenyl group (or diphenylaminophenyl substituent for a related molecule), TBBN (Figure 11), has been synthesized in several steps, and its fluorescence behavior was attributed to the through-space intramolecular charge transfer from the dimethylamino group to the dimesitylboryl group ($\lambda_{\text{abs}} = 315$ and 319 nm ; $\lambda_{\text{em}} = 445$ and 470 nm); binding of the TBBN–boron to fluoride or cyanide anions results in the ~ 72 – 140 nm blue shift and is accompanied by visual color changes.⁵⁹ The binding of fluoride to TBBN results a color change from colorless to orange, while binding of the cyanide anion results in color change

from colorless to green. Thus these molecules serve as potential colorimetric sensors for the fluoride (as well as cyanide) anions. Fluorescent sensors based on dimesitylborylthiophene derivatives have been recently reported.⁶⁰

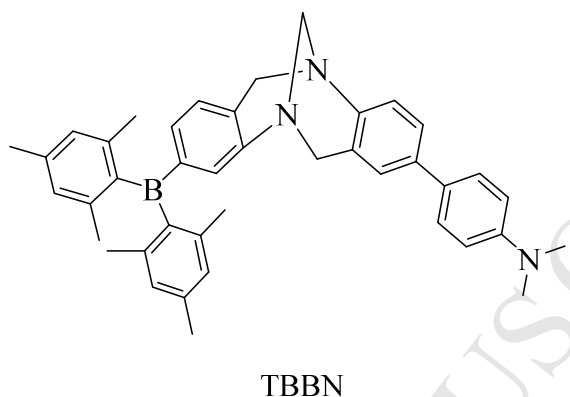


Figure 11. Structure of a colorimetric fluoride anion sensor, TBBN

5. Arylboronic acids as colorimetric sensors of fluoride anion

Arylboronic acids typically bind through hydrogen-bonding interactions with soft anions, such as Cl^- , Br^- , HSO_4^- , and AcO^- , whereas they bind through Lewis acid–base type of interactions with hard anions, such as F^- and H_2PO_4^- . Even in the latter cases, the anion adducts are partly stabilized through hydrogen–bonding interactions involving the hydroxyl groups of the boronic acids, and there may be equilibrating structures involving the hydrogen–bonded adducts with the boron–fluoride adducts, as shown in Figure 12.⁶¹ The hydrogen-bonding interactions of the arylboronic acids are relatively insensitive to the solvent polarity, and are comparable to those of the various urea derivatives. The Bronsted acid–base binding constant of arylboronic acids with acetate anions in DMSO-d_6 range from 990 M^{-1} for the 4-methoxyphenylboronic acid to 6200 M^{-1} for the 4-nitrophenylboronic acid. Appropriately chromophore-substituted derivatives of these compounds may afford the colorimetric sensors of fluoride anions in aqueous media (vide infra).

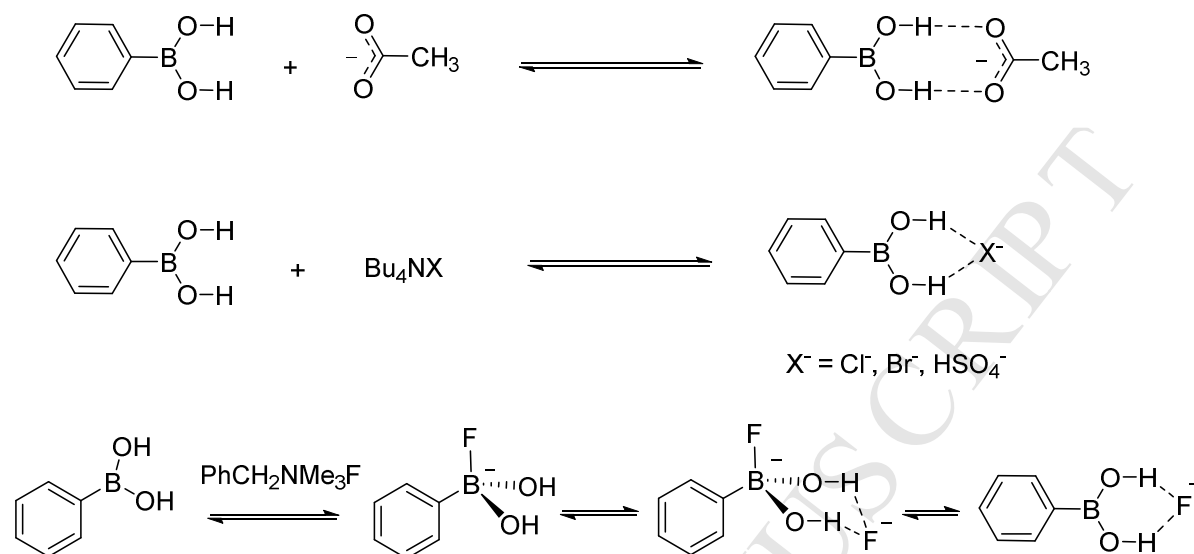
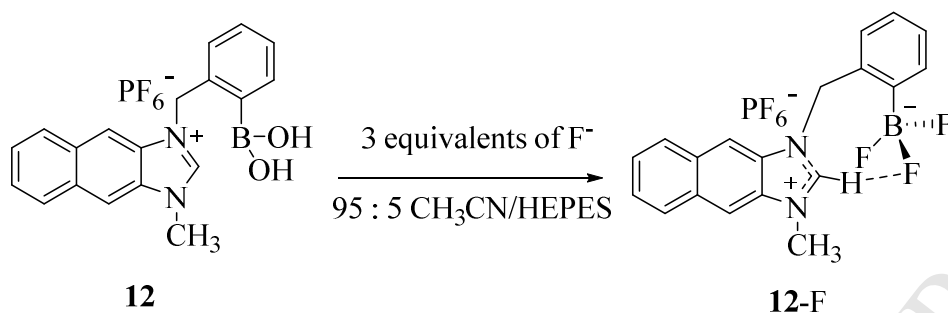


Figure 12. Arylboronic acids as anion receptors; hydrogen-bonding interactions are involved in binding to the soft anions such as chloride, bromide, hydrogen sulfate, and acetate anions, whereas Lewis acid–base type of binding is observed for the fluoride anion. However, in the latter case, there may be a minor equilibrium with the hydrogen bonded structures.

5.1. Bidentate boronic acid derived fluoride anion sensors

A bidentate N-(arylnaphthyl)imidazolium boronic acid derived fluoride anion sensor **12** involving the intramolecular CH-anion interaction has been reported by Xu and coworkers.⁶² In aqueous acetonitrile consisting of HEPES buffer the cationic boronic acid **12** reacts with three equivalents of the fluoride anion to give the corresponding trifluoroborate (**12-F**), which exhibits C–H...F hydrogen bonding interaction with one of the B–F fluorines and thereby shows a ratiometric fluorescent behavior. The corresponding *meta*- and *para*- boronic acid substituted versions of **12** failed to show the fluoride anion binding affinity. The involvement of the CH...F hydrogen bonding interactions was confirmed by ¹⁹F NMR spectroscopy.



6. Conclusions and outlook

Selective fluoride anion receptors would substantially impact the area of lithium ion batteries, and in this context there has been growing interest in the design and electrochemical studies of the boron based anion receptors. Similarly, selective fluoride anion receptors in aqueous media would substantially impact as chemosensors in the areas of medicinal chemistry and environmental chemistry, and as phase transfer catalysts in organic synthesis. It is hoped that further studies on delineating the factors controlling the fluoride anion selectivity, as well as the studies on modulating the reversibility of the fluoride anion binding, would significantly impact the areas of lithium ion (and lithium-air) batteries and fluoride-ion specific chemosensors.

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